Lab Sheet 2

Tasks

1. Create a script.js file. In it, define a JavaScript array of product objects (id, name, price, image) representing your store's catalog — at least 8 products.
2. On products.html, use JavaScript to dynamically render the product cards from your array into the page (instead of hardcoded HTML blocks) using DOM methods like createElement / innerHTML with template literals.
3. Implement an 'Add to Cart' click handler on each product card that adds the product to a cart array (store it in a JS variable, and also persist it using localStorage so it survives a page refresh).
4. On cart.html, read the cart from localStorage and dynamically render each item with a working quantity input and a 'Remove' button.
5. Write a function calculateTotal() that recalculates and displays the cart's
grand total whenever quantity changes or an item is removed.
6. On checkout.html, add form validation using JavaScript: Name and Address must not be empty, Pincode must be exactly 6 digits, Phone must be exactly 10 digits. Show inline error messages instead of using alert().
7. On successful validation, show a simple confirmation message on the page (e.g., 'Order placed!') and clear the cart from localStorage.
8. Add a small script on every page to update a 'cart item count' badge next to the Cart link in the navbar, read from localStorage.

Solutions-> index.html

```
<! DOCTYPE html >
<html lang ="en" >
<head >
      <meta charset ="UTF-8" >
      <meta name ="viewport"   content ="width=device-width,
initial-scale=1.0"  >
      <title >Technical Store  </title >
      <link  rel ="stylesheet"  href  ="style.css" >
</head >
<body >
<nav >
      <ahref  ="index.html"   >Home </a>
      <ahref  ="products.html"   >Products </a>
      <ahref  ="cart.html"  >Cart  <span id ="cart-count"  >0</span ></a>
      <ahref  ="checkout.html"   >Checkout </a>
</nav >
<header >
      <h1 >Welcome to Technical Store   </h1 >
      <p >Your one-stop shop for quality tech products.     </p >
```

```
        <ahref ="products.html"  >Shop Now </a>
</header >
<section >
        <h2>Featured Products  </h2>
        <p>Explore our latest collection of technology products.       </p>
        <ahref ="products.html"  >View All Products  </a>
</section >
<script src="script.js"  ></script >
</body >
</html >
```

products.html

```
<! DOCTYPE html >
<html lang ="en" >
<head >
        <meta charset ="UTF-8" >
        <meta name ="viewport"  content ="width=device-width,
initial-scale=1.0"  >
        <title >Technical Store - Products  </title >
        <link  rel ="stylesheet"  href ="style.css" >
</head >
<body >
<nav >
        <ahref ="index.html"  >Home </a>
        <ahref ="products.html"  >Products </a>
        <ahref ="cart.html"  >Cart  <span id ="cart-count"  >0</span ></a>
        <ahref ="checkout.html"  >Checkout </a>
</nav >
<h1 >Our Products  </h1 >
<div id ="product-container"    ></div >
<script src="script.js"  ></script >
</body >
</html >
```

checkout.html

```
<! DOCTYPE html >
<html lang ="en" >
<head >
        <meta charset ="UTF-8" >
        <meta name ="viewport"  content ="width=device-width,
initial-scale=1.0"  >
        <title >Technical Store - Checkout  </title >
        <link  rel ="stylesheet"  href ="style.css" >
</head >
<body >
```

```
<nav >
      <ahref ="index.html"  >Home </a>
      <ahref ="products.html"  >Products </a>
      <ahref ="cart.html"  >Cart  <span id ="cart-count"  >0</span ></a>
      <ahref ="checkout.html"  >Checkout </a>
</nav >
<h1 >Checkout </h1 >
<form id ="checkout-form"  >
      <label >Name </label >
      <input type ="text" id ="name"  >
      <span id ="name-error"  ></span >
      <label >Address </label >
      <textarea  id ="address" ></textarea >
      <span id ="address-error"  ></span >
      <label >Pincode </label >
      <input type ="text" id ="pincode"  maxlength  ="6" >
      <span id ="pincode-error"  ></span >
      <label >Phone </label >
      <input type ="text" id ="phone"  maxlength  ="10" >
      <span id ="phone-error"  ></span >
      <button type ="submit" >Place Order  </button >
</form >
<h2 id ="confirmation"  ></h2 >
<script src ="script.js" ></script >
</body >
</html >
```

cart.html

```
<! DOCTYPE html >
<html lang ="en" >
<head >
      <meta charset ="UTF-8" >
      <meta name ="viewport"  content ="width=device-width,
initial-scale=1.0"  >
      <title >Technical Store - Cart  </title >
      <link  rel ="stylesheet"  href ="style.css" >
</head >
<body >
<nav >
      <ahref ="index.html"  >Home </a>
      <ahref ="products.html"  >Products </a>
      <ahref ="cart.html"  >Cart  <span id ="cart-count"  >0</span ></a>
      <ahref ="checkout.html"  >Checkout </a>
</nav >
```

```
<h1>Shopping Cart</h1>
<div id="cart-container"></div>
<h2>Grand Total: <span id="cart-total">₹0</span></h2>
<a href="checkout.html">Proceed to Checkout</a>
<script src="script.js"></script>
</body>
</html>
```

style.css

```
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}
body{
    font-family:Arial,sans-serif;
    background:#f5f7fa;
    color:#222;
}
nav{
    background:#1f2937;
    padding:18px 40px;
    display:flex;
    align-items:center;
    gap:30px;
}
nav a{
    color:white;
    text-decoration:none;
    font-size:17px;
}
nav a:hover{
    color:#38bdf8;
}
#cart-count{
    background:#ef4444;
    color:white;
    padding:3px 7px;
    border-radius:50%;
    font-size:12px;
    margin-left:3px;
}
header{
    text-align:center;
```

```
        padding :100px 20px;
        background :linear-gradient (135deg,#dbeafe,#eff6ff);
}
header h1{
        font-size :48px;
        margin-bottom :20px;
}
header p{
        font-size :20px;
        margin-bottom :30px;
        color :#555;
}
header a{
        display :inline-block ;
        padding :13px 28px;
        background :#2563eb;
        color :white ;
        text-decoration :none;
        border-radius :6px;
        font-size :17px;
}
header a:hover {
        background :#1d4ed8;
}
section{
        text-align :center ;
        padding :60px 20px;
}
section h2{
        font-size :32px;
        margin-bottom :15px;
}
section p{
        color :#666;
        font-size :17px;
        margin-bottom :25px;
}
section a{
        display :inline-block ;
        padding :10px 20px;
        background :#111827;
        color :white ;
        text-decoration :none;
```

```
        border-radius : 5px;
}
section a:hover {
        background : #374151;
}
#product-container {
        max-width : 1200px;
        margin : 40px auto;
        padding : 020px;
        display : grid;
        grid-template-columns : repeat (4,1fr);
        gap: 25px;
}
.product-card {
        background : white;
        border-radius : 10px;
        padding : 15px;
        box-shadow : 03px 12px rgba (0,0,0,0.1);
        text-align : center;
        transition :transform 0.2s;
}
.product-card:hover {
        transform : translateY (-5px);
}
.product-card img{
        width : 100%;
        height : 200px;
        object-fit : cover;
        border-radius : 8px;
}
.product-card h3{
        margin : 15px 010px;
}
.product-card p{
        font-size : 20px;
        font-weight : bold;
        margin-bottom : 15px;
}
.product-card button {
        background : #2563eb;
        color : white;
        border : none;
        padding : 10px 18px;
```

```
        border-radius :5px;
        cursor :pointer ;
}
.product-card button:hover {
        background :#1d4ed8;
}
#cart-container {
        max-width :900px;
        margin :40px auto;
        padding :020px;
}
.cart-item {
        display :flex ;
        align-items :center ;
        gap:25px;
        background :white ;
        padding :20px;
        margin-bottom :20px;
        border-radius :10px;
        box-shadow :03px 10px rgba (0,0,0,0.1);
}
.cart-item img{
        width :140px;
        height :120px;
        object-fit :cover ;
        border-radius :8px;
}
.cart-item input {
        width :60px;
        padding :8px;
        margin :10px;
}
.cart-item button {
        background :#dc2626;
        color :white ;
        border :none ;
        padding :8px 15px;
        border-radius :5px;
        cursor :pointer ;
}
.cart-item button:hover {
        background :#b91c1c;
}
```

```
#cart-total {
        font-weight :bold ;
        color :#2563eb;
}
form {
        max-width :500px ;
        margin :40px auto ;
        padding :30px ;
        background :white ;
        border-radius :10px ;
        box-shadow :03px 15px rgba (0,0,0,0.1);
        display :flex ;
        flex-direction :column ;
}
form label {
        margin-top :10px ;
        margin-bottom :5px ;
        font-weight :bold ;
}
form input ,
form textarea {
        padding :10px ;
        border :1px solid #ccc;
        border-radius :5px ;
        font-size :15px ;
}
form textarea {
        min-height :100px ;
        resize :vertical ;
}
form span{
        color :#dc2626;
        font-size :14px ;
        margin-top :3px ;
}
form button {
        margin-top :20px ;
        padding :12px ;
        background :#2563eb;
        color :white ;
        border :none ;
        border-radius :5px ;
        cursor :pointer ;
```

```
        font-size : 16px ;
}
form button:hover  {
        background  : #1d4ed8;
}
#confirmation  {
        text-align  : center ;
        color : green ;
        margin  : 20px ;
}
@media (max-width  : 900px ){
        #product-container  {
                grid-template-columns  : repeat (2,1fr );
        }
}
@media (max-width  : 600px ){
        nav {
                padding : 15px ;
                gap: 15px ;
                flex-wrap  : wrap ;
        }
        #product-container  {
                grid-template-columns  : 1fr ;
        }
        .cart-item {
                flex-direction  : column ;
                text-align  : center ;
        }
        header h1{
                font-size : 35px ;
        }
}
```

script.js

```
const products =[
{id: 1,name: "Wireless
Headphones" ,price: 1499,image: "https://images.unsplash.com/photo-1505740
420928-5e560c06d30e" },
{id: 2,name: "Smart
Watch" ,price: 2499,image: "https://images.unsplash.com/photo-152327533568
4-37898b6baf30" },
{id: 3,name: "Running
Shoes" ,price: 1999,image: "https://images.unsplash.com/photo-1542291026-7
eec264c27ff" },
```

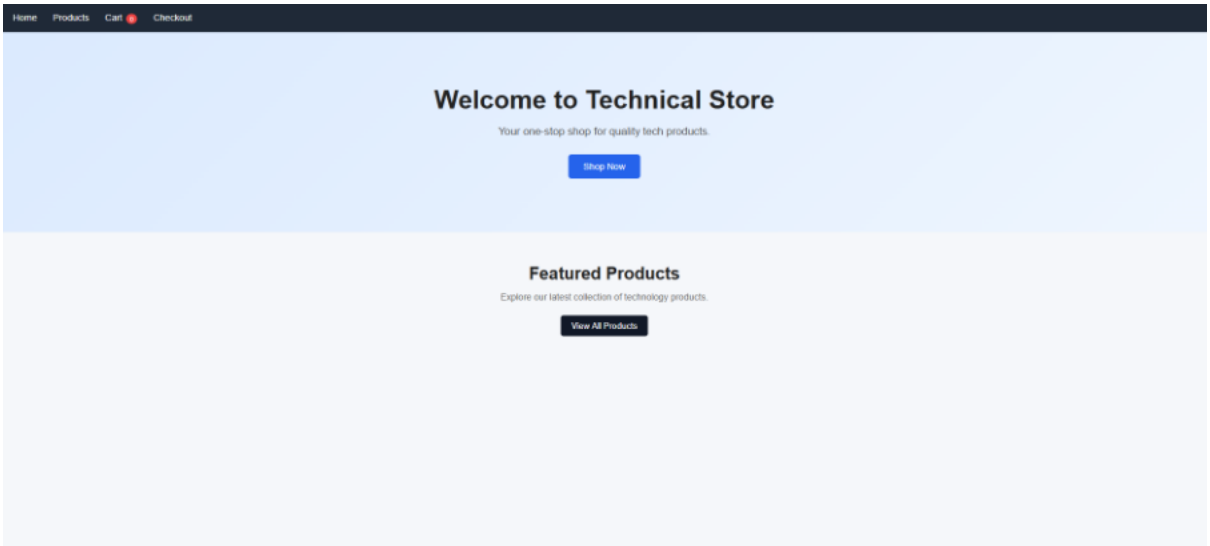
```
{id: 4,name: "Laptop
Backpack" ,price: 1299,image: "https://images.unsplash.com/photo-155306240
7-98eeb64c6a62"},
{id: 5,name: "Bluetooth
Speaker" ,price: 1799,image: "https://images.unsplash.com/photo-1608043152
269-423dbba4e7e1" },
{id: 6,name: "Gaming
Mouse" ,price: 999,image: "https://images.unsplash.com/photo-1527814050087
-3793815479db" },
{id: 7,name: "Mechanical
Keyboard" ,price: 2299,image: "https://images.unsplash.com/photo-158782974
1301-dc798b83add3" },
{id: 8,name: "Sunglasses" ,price: 799,image: "https://images.unsplash.com/ph
oto-1511499767150-a48a237f0083" }
];
let cart =JSON.parse (localStorage .getItem ("cart" ))|| [];
function updateCartCount (){
    const count =document. getElementById ("cart-count" );
    if (count){
        const
totalItems =cart. reduce ((total ,item )=>total +item.quantity, 0);
        count.textContent =totalItems;
    }
}
function saveCart (){
    localStorage. setItem ("cart" ,JSON.stringify (cart));
    updateCartCount ();
}
function addToCart (id ){
    const product =products. find (product =>product.id ===id);
    const existingItem =cart. find (item =>item.id ===id);
    if (existingItem){
        existingItem.quantity ++;
    }else{
        cart. push ({
            id: product.id,
            name: product.name,
            price: product.price,
            image: product.image,
            quantity: 1
        });
    }
    saveCart ();
```

```
}
function renderProducts (){
    const container =document. getElementById ("product-container" );
    if(!container) return ;
    container.innerHTML ="";
    products. forEach (product =>{
        const card =document. createElement ("div" );
        card.className ="product-card" ;
        card.innerHTML =`
            <img src=" ${product .image}" alt=" ${product .name}">
            <h3> ${product .name}</h3>
            <p>₹${product .price }</p>
            <button onclick="addToCart( ${product .id})">Add to
Cart</button>
        `;
        container. appendChild (card);
    });
}
function renderCart (){
    const container =document. getElementById ("cart-container" );
    const totalElement =document. getElementById ("cart-total" );
    if(!container) return ;
    container.innerHTML ="";
    if(cart. length ===0){
        container.innerHTML ="<p>Your cart is empty.</p>" ;
        if(totalElement){
            totalElement.textContent ="₹0";
        }
        return ;
    }
    cart. forEach (item =>{
        const cartItem =document. createElement ("div" );
        cartItem.className ="cart-item" ;
        cartItem.innerHTML =`
            <img src=" ${item .image}" alt=" ${item .name}">
            <div>
                <h3> ${item .name}</h3>
                <p>Price: ₹${item .price }</p>
                <label>Quantity:</label>
                <input type="number" min="1" value=" ${item .quantity }"
onchange="updateQuantity( ${item .id},this.value)">
                <button
onclick="removeFromCart( ${item .id})">Remove</button>
```

```
                </div>
            `;
            container.appendChild(cartItem);
        });
        calculateTotal();
}
function updateQuantity(id,quantity){
        const item =cart.find(item =>item.id ===id);
        if(item){
            item.quantity =Math.max(1,parseInt(quantity) || 1);
            saveCart();
            renderCart();
        }
}
function removeFromCart(id){
        cart =cart.filter(item =>item.id !== id);
        saveCart();
        renderCart();
}
function calculateTotal(){
        const totalElement =document.getElementById("cart-total");
        if(!totalElement) return;
        const total =cart.reduce((sum,item)=>{
            return sum +(item.price *item.quantity);
        },0);
        totalElement.textContent =`₹${total}`;
}
function validateCheckout(event){
        event.preventDefault();
        const name =document.getElementById("name");
        const address =document.getElementById("address");
        const pincode =document.getElementById("pincode");
        const phone =document.getElementById("phone");
        const nameError =document.getElementById("name-error");
        const addressError =document.getElementById("address-error");
        const pincodeError =document.getElementById("pincode-error");
        const phoneError =document.getElementById("phone-error");
        const confirmation =document.getElementById("confirmation");
        nameError.textContent ="";
        addressError.textContent ="";
        pincodeError.textContent ="";
        phoneError.textContent ="";
        confirmation.textContent ="";
```

```
    let valid =true;
    if(name.value. trim ()===""){
        nameError.textContent ="Name is required." ;
        valid =false;
    }
    if(address.value. trim ()===""){
        addressError.textContent ="Address is required." ;
        valid =false;
    }
    if(!/^\d{6}$/.test(pincode.value)){
        pincodeError.textContent ="Pincode must be exactly 6 digits." ;
        valid =false;
    }
    if(!/^\d{10}$/.test(phone.value)){
        phoneError.textContent ="Phone must be exactly 10 digits." ;
        valid =false;
    }
    if(valid){
        confirmation.textContent ="Order placed!" ;
        cart =[];
        localStorage. removeItem ("cart");
         updateCartCount ();
        document. getElementById ("checkout-form").reset();
    }
}
document .addEventListener ("DOMContentLoaded" ,()=>{
    updateCartCount ();
    renderProducts ();
    renderCart ();
    constcheckoutForm =document. getElementById ("checkout-form");
    if(checkoutForm){
        checkoutForm. addEventListener ("submit" ,validateCheckout);
    }
});
```

Output ->


Home
Products
Cart 0
Checkout
Welcome to Technical Store
Your one-stop shop for quality tech products.
Shop Now
Featured Products
Explore our latest collection of technology products.
View All Products